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Safety data sheet

BASF Safety data sheet according to UN GHS 4th rev.
Date / Revised: 09.04.2015 Version: 1.0
Product: **Beethoven TR Total Release Miticide/Insecticide**
(ID no. 30733044/SDS_CPA_00/EN)
Date of print 09.07.2025

1. Identification

Product identifier

Beethoven TR Total Release Miticide/Insecticide

Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: crop protection product, insecticide

Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Crop Protection

Telephone: +49 621 60-27777
E-mail address: Produktinformation-Pflanzenschutz@basf.com

Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

STOT SE 3 (Vapours may cause drowsiness and dizziness.)
Aquatic Acute 1
Aquatic Chronic 1
Aerosols 1

For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System (GHS)

Pictogram:

Signal Word:
Danger

Hazard Statement:

H222	Extremely flammable aerosol.
H336	May cause drowsiness or dizziness.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.
H229	Pressurized container: May burst if heated.

Precautionary Statements (Prevention):

P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P271	Use only outdoors or in a well-ventilated area.
P261	Avoid breathing spray.
P211	Do not spray on an open flame or other ignition source.
P251	Do not pierce or burn, even after use.

Precautionary Statements (Response):

P312	Call a POISON CENTER or doctor/physician if you feel unwell.
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P391	Collect spillage.

Precautionary Statements (Storage):

P403 + P233	Store in a well-ventilated place. Keep container tightly closed.
P410 + P412	Protect from sunlight. Do no expose to temperatures exceeding 50°C/ 122°F.
P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents/container to hazardous or special waste collection point.

Labeling of special preparations (GHS):
Repeated exposure may cause skin dryness or cracking.

Only for professional users.

According to UN GHS criteria

Hazard determining component(s) for labelling: etoxazol, ACETONE/DIMETHYLKETONE, dimethyl ether

Other hazards

According to UN GHS criteria

See section 12 - Results of PBT and vPvB assessment.

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

crop protection product, insecticide

Hazardous ingredients (GHS)
According to UN GHS criteria

etoxazol

Content (W/W): 5,2 %	Aquatic Acute 1
CAS Number: 153233-91-1	Aquatic Chronic 1
	H400, H410

acetone; propan-2-one; propanone

Content (W/W): < 35 %	Flam. Liq. 2
CAS Number: 67-64-1	Eye Dam./Irrit. 2A
EC-Number: 200-662-2	STOT SE 3 (drowsiness and dizziness)
INDEX-Number: 606-001-00-8	H225, H319, H336
	EUH066
	EUH066

dimethyl ether

Content (W/W): < 65 %	Press. Gas Liquef. Gas
CAS Number: 115-10-6	Flam. gases (incl. chem. unstable gases) 1
EC-Number: 204-065-8	H280, H220

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water.

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: The most important known symptoms and effects are described in the labelling (see section 2) and/or in section 11., Further important symptoms and effects are so far not known.

Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
foam, dry powder, carbon dioxide, water spray

Special hazards arising from the substance or mixture

Carbon dioxide, carbon monoxide
The substances/groups of substances mentioned can be released in case of fire.

Advice for fire-fighters

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations. In case of fire and/or explosion do not breathe fumes. Keep containers cool by spraying with water if exposed to fire.

6. Accidental Release Measures

Personal precautions, protective equipment and emergency procedures
Do not breathe vapour/spray. Use personal protective clothing. Avoid contact with the skin, eyes and clothing.

Environmental precautions
Do not discharge into the subsoil/soil. Do not discharge into drains/surface waters/groundwater.

Methods and material for containment and cleaning up
For small amounts: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr).
For large amounts: Dike spillage. Pump off product.
Dispose of absorbed material in accordance with regulations. Collect waste in suitable containers, which can be labeled and sealed. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations.

Reference to other sections
Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

7. Handling and Storage

Precautions for safe handling
No special measures necessary if stored and handled correctly. Ensure thorough ventilation of stores and work areas. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.

Conditions for safe storage, including any incompatibilities
Segregate from foods and animal feeds.

Suitable materials for containers: High density polyethylene (HDPE), Low density polyethylene (LDPE)
Further information on storage conditions: Keep away from heat. Protect from direct sunlight.

Specific end use(s)
For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

67-64-1: acetone; propan-2-one; propanone
115-10-6: dimethyl ether

Exposure controls

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for higher concentrations or long-term effect: Combination filter for gases/vapours of organic, inorganic, acid inorganic, alkaline compounds and toxic particles (e. g. EN 14387 Type ABEK-P3)

Hand protection:
Suitable chemical resistant safety gloves (EN 374) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN 374): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
Wearing of closed work clothing is recommended. The statements on personal protective equipment in the instructions for use apply when handling crop-protection agents in final-consumer packing. Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

9. Physical and Chemical Properties

Information on basic physical and chemical properties

Form:	aerosol
Colour:	pale straw yellow
Odour:	characteristic, of acetone
Odour threshold:	Not determined due to potential health hazard by inhalation.

pH value:	approx. 5,5 - 7,5 (10 g/l, 22,5 °C)	
Melting point:	approx. -95 °C Information applies to the solvent.	
Boiling point:	approx. 56 °C Information applies to the solvent.	
Flash point:	approx. -18 °C The statements are based on the properties of the individual components.	
Evaporation rate:	not applicable	
Flammability:	Extremely flammable.	
Flammability of Aerosol Products:	> 18 in no flashback	(ASTM D 3065)
Lower explosion limit:	2,4 %(V) (20 °C, 1013 hPa) Information applies to the solvent.	
Upper explosion limit:	14,3 %(V) (100 °C, 1013 hPa) Information applies to the solvent.	
Ignition temperature:	approx. 535 °C Information applies to the solvent.	
Vapour pressure:	approx. 246 hPa (20 °C) Information applies to the solvent.	
Density:	approx. 0,82 g/cm3 (20 °C)	
Relative vapour density (air):	not applicable	
Solubility in water:	emulsifiable	

Information on: acetone; propan-2-one; propanone

Partitioning coefficient n-octanol/water (log Kow):	*-0,24* *(25 °C)*	*(Calculation Hansch/Leo)*

Information on: dimethyl ether

Partitioning coefficient n-octanol/water (log Kow):	*-0,18*	*(calculated)*

Thermal decomposition:	No decomposition if stored and handled as prescribed/indicated.
Viscosity, dynamic:	0,52 mPa.s (18,5 °C)
Explosion hazard:	Based on the chemical structure there is no indicating of explosive properties., The product has not been tested. The statement has been derived from the properties of the individual components.
Fire promoting properties:	not fire-propagating

Other information

Other Information:

If necessary, information on other physical and chemical parameters is indicated in this section.

10. Stability and Reactivity

Reactivity
No hazardous reactions if stored and handled as prescribed/indicated.

Chemical stability
The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions
No hazardous reactions if stored and handled as prescribed/indicated.

Conditions to avoid
See MSDS section 7 - Handling and storage.

Incompatible materials

Substances to avoid:
strong acids, strong bases, strong oxidizing agents

Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Virtually nontoxic after a single ingestion. Virtually nontoxic by inhalation. Virtually nontoxic after a single skin contact.

Experimental/calculated data:
LD50 rat (oral): > 5.000 mg/kg

LC50 rat (by inhalation): > 2,07 mg/l
No mortality was observed. An aerosol with respirable particles was tested.

LD50 rat (dermal): > 5.000 mg/kg
No mortality was observed.

Irritation

Assessment of irritating effects:
Not irritating to the skin. Not irritating to the eyes.

Experimental/calculated data:
Skin corrosion/irritation rabbit: non-irritant

Serious eye damage/irritation rabbit: non-irritant

Respiratory/Skin sensitization

Assessment of sensitization:
There is no evidence of a skin-sensitizing potential.

Experimental/calculated data:
Buehler test guinea pig: Non-sensitizing.

Germ cell mutagenicity

Assessment of mutagenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. Mutagenicity tests revealed no genotoxic potential.

Carcinogenicity

Assessment of carcinogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. The results of various animal studies gave no indication of a carcinogenic effect.

Reproductive toxicity

Assessment of reproduction toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: acetone; propan-2-one; propanone
Assessment of reproduction toxicity:
As shown in animal studies, the product may cause damage to the testes after repeated high exposures that cause other toxic effects.

Developmental toxicity

Assessment of teratogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. Animal studies gave no indication of a developmental toxic effect at doses that were not toxic to the parental animals.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Possible narcotic effects (drowsiness or dizziness).

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: acetone; propan-2-one; propanone
Assessment of repeated dose toxicity:
The substance may cause damage to the testes after repeated ingestion of high doses, as shown in animal studies. The substance may cause damage to the hematological system after repeated ingestion of high doses. The substance may cause damage to the kidney after repeated ingestion of high doses, as shown in animal studies.

Other relevant toxicity information

Misuse can be harmful to health.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components. Very toxic to aquatic organisms. May cause long-term adverse effects in the aquatic environment.

Information on: etoxazol
Toxicity to fish:
LC50 2,8 mg/l, Oncorhynchus mykiss

LC50 1,4 mg/l, Lepomis macrochirus

Information on: etoxazol
Aquatic invertebrates:
EC50 (48 h) 0,0071 mg/l, Daphnia magna

LC50 (96 h) 0,0012 mg/l, Crassostrea virginica

Information on: etoxazol
Aquatic plants:
EC50 (72 h) > 10 mg/l (biomass), Pseudokirchneriella subcapitata

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Information on: etoxazol
Chronic toxicity to fish:
No observed effect concentration (89 d) 0,015 mg/l, Oncorhynchus mykiss (OECD Guideline draft)

Information on: etoxazol
Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d) 0,0002 mg/l, Daphnia magna
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Persistence and degradability

Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria). The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: etoxazol
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria).

Bioaccumulative potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: acetone; propan-2-one; propanone
Assessment bioaccumulation potential:
Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

Information on: dimethyl ether
Bioaccumulation potential:
Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

Mobility in soil

Assessment transport between environmental compartments:
Volatility: The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: acetone; propan-2-one; propanone
Assessment transport between environmental compartments:
Volatility: The substance will slowly evaporate into the atmosphere from the water surface.

Adsorption in soil: Adsorption to solid soil phase is not expected.

Results of PBT and vPvB assessment

The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

13. Disposal Considerations

Waste treatment methods

Must be sent to a suitable incineration plant, observing local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

14. Transport Information

Land transport

ADR

UN number	UN1950
UN proper shipping name:	AEROSOLS (contains DIMETHYLETHER, ETOXAZOL)
Transport hazard class(es):	2.1, EHSM
Packing group:	Not applicable
Environmental hazards:	yes
Special precautions for user:	Tunnel code: D

RID

UN number	UN1950
UN proper shipping name:	AEROSOLS (contains DIMETHYLETHER, ETOXAZOL)
Transport hazard class(es):	2.1, EHSM

Packing group:	Not applicable
Environmental hazards:	yes
Special precautions for user:	None known

Inland waterway transport

ADN

UN number	UN1950
UN proper shipping name:	AEROSOLS (contains DIMETHYLETHER, ETOXAZOL)
Transport hazard class(es):	2.1, EHSM
Packing group:	Not applicable
Environmental hazards:	yes
Special precautions for user:	None known
Transport in inland waterway vessel:	

Sea transport

IMDG

UN number:	UN 1950
UN proper shipping name:	AEROSOLS (contains DIMETHYLETHER, ETOXAZOL)
Transport hazard class(es):	2.1, EHSM
Packing group:	Not applicable
Environmental hazards:	yes Marine pollutant: YES
Special precautions for user:	None known

Air transport

IATA/ICAO

UN number:	UN 1950
UN proper shipping name:	AEROSOLS, FLAMMABLE (contains DIMETHYLETHER, ETOXAZOL)
Transport hazard class(es):	2.1
Packing group:	Not applicable
Environmental hazards:	No Mark as dangerous for the environment is needed
Special precautions for user:	None known

UN number

See corresponding entries for “UN number” for the respective regulations in the tables above.

UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

Environmental hazards
See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

Special precautions for user
See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code

Regulation:	Not evaluated
Shipment approved:	Not evaluated
Pollution name:	Not evaluated
Pollution category:	Not evaluated
Ship Type:	Not evaluated

15. Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

Not applicable

16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

STOT SE	Specific target organ toxicity — single exposure
Aquatic Acute	Hazardous to the aquatic environment - acute
Aquatic Chronic	Hazardous to the aquatic environment - chronic
Aerosols	Aerosols
Flam. Liq.	Flammable liquids
Eye Dam./Irrit.	Serious eye damage/eye irritation
Press. Gas	Gases under pressure
Flam. gases (incl. chem. unstable gases)	Flammable gases (including chemically unstable gases)
H400	Very toxic to aquatic life.

H410	Very toxic to aquatic life with long lasting effects.
H225	Highly flammable liquid and vapour.
H319	Causes serious eye irritation.
H336	May cause drowsiness or dizziness.
H280	Contains gas under pressure; may explode if heated.
H220	Extremely flammable gas.
EUH066	Repeated exposure may cause skin dryness or cracking.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. The data do not describe the product's properties (product specification). Neither should any agreed property nor the suitability of the product for any specific purpose be deduced from the data contained in the safety data sheet. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

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